

Harthik Royal Mallichetty

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PROFESSIONAL SUMMARY

AI Engineer and May 2026 MSBA graduate with production experience building end-to-end AI systems. Built Signal, a financial intelligence platform covering 70 S&P 500 companies with a 10-step RAG pipeline, Voyage AI finance embeddings, pgvector HNSW retrieval, and GPT-4o-mini on GCP. Previously built a production ML system at Prolifics used by 200 developers daily, still running in 2025. Proficient in Python, LangChain, FastAPI, dbt, Airflow, PostgreSQL, GCP, and React.

EDUCATION

The University of Texas at Dallas, Richardson, Texas

Master of Science in Business Analytics and Artificial Intelligence – Scholar With Distinction

May 2026

GPA 3.81

GITAM University, Hyderabad, India

April 2024

Bachelor of Technology in Computer Science and Business Systems

GPA 3.73

PROFESSIONAL EXPERIENCE

Prolifics – Data & Analytics Intern | Hyderabad, India

Feb 2023 – Jan 2024

- Identified need for sprint outcome prediction and built SmartQE system solo: designed SQL pipeline on 100K+ PostgreSQL records, trained TensorFlow regression model achieving 89%+ accuracy, deployed Flask REST API into company portal; 200 developers use it daily and it is still running in production 2025.
- Automated ETL workflows processing 50K+ rows/day and built Power BI dashboards tracking KPIs for Finance, DevOps, and QA; reduced reporting cycle time by 40% and improved downstream accuracy by 25%.

HeiDigi – Digital Marketing Analyst Intern | Hyderabad, India

Mar 2022 – Aug 2022

- Built automated Tableau dashboards pulling Google Ads performance data for hotel client campaigns, surfacing KPIs that improved ROI by 20% and CTR by 18%.
- Designed and executed A/B tests applying hypothesis testing and statistical inference; translated findings into recommendations that drove 15% revenue impact increase.

PROJECTS

Signal – Financial Intelligence | Python, FastAPI, pgvector, dbt, Airflow, Voyage AI, GPT-4o-mini, GCP

May 2026

- Architected 5-layer production system covering 70 S&P 500 companies: 8-module Python ingestion pipeline pulling SEC EDGAR filings and Polygon.io market data, XBRL parser with bank-specific fallback logic, dbt 3-layer transformation computing 65 metrics per company, and FastAPI on GCP Cloud Run with API key auth across 11 endpoints.
- Built 10-step RAG pipeline with GPT-4o-mini query classification and ticker extraction, Voyage AI voyage-finance-2 embeddings, pgvector HNSW cosine retrieval, 6-exchange conversation memory, and LangSmith observability; guardrail check runs before every LLM call to eliminate wasted spend.

Quant Risk Desk | Python, SQL, dbt, Apache Airflow, PostgreSQL, Docker, Tableau

April 2026

- Built automated pipeline via 9-task Airflow DAG ingesting 21K+ daily records across 11 S&P 500 sector ETFs; implemented 18 dbt data quality tests with zero violations and 3-layer transformation computing VaR/CVaR at 95%/99%, Sharpe/Sortino ratios, rolling Beta, and Macro classification using Fed Funds Rate, CPI, VIX, and yield curve signals.
- Identified diversification collapse from 0.47 to 0.88 correlation during March 2020 COVID crash across 182K+ pair-day observations; delivered Tableau dashboard with sector KPI scorecards, VaR envelopes, and correlation heatmaps.

Global Supply Chain Intelligence | Python, SQL, XGBoost, LightGBM, SHAP, NetworkX, Plotly

Mar 2026

- Built CUSUM and Mahalanobis anomaly detection achieving 100% disruption recall with 8-week lead time across 78K weekly records and 500 SKUs; trained XGBoost and LightGBM ensemble achieving Recall 70.9% with SHAP TreeExplainer attributing 59% of model weight to disruption signals.
- Executed Monte Carlo simulation with 10K runs per SKU across 3 geopolitical stress scenarios (baseline \$13.8M, moderate \$18.1M, severe \$19.1M); delivered 6-tab interactive Plotly dashboard for non-technical stakeholder review.

SKILLS

AI and LLM: RAG Pipeline Design, LangChain, LangGraph, pgvector HNSW, Voyage AI (voyage-finance-2), OpenAI API (GPT-4o-mini), Claude API, Gemini API, Groq, Prompt Engineering, Guardrail Design, LangSmith

Programming: Python (Pandas, NumPy, scikit-learn, XGBoost, LightGBM, TensorFlow, PyTorch, SHAP, SciPy, Plotly), SQL (CTEs, Window Functions, Rolling Aggregations, Query Optimization), TypeScript, R

Data Engineering: Apache Airflow (DAGs), dbt (Staging/Intermediate/Marts, custom tests), FastAPI, Flask, PostgreSQL, pgvector, DuckDB, Docker, ETL Automation, Dimensional Modeling

Cloud and Tools: GCP (Cloud Run, Cloud SQL), Vercel, GitHub Actions, React 18, Vite, TailwindCSS, Framer Motion, Zustand, TanStack Query, TradingView Lightweight Charts, n8n, SEC EDGAR API, Polygon.io

ML and Statistics: XGBoost, LightGBM, TensorFlow, PyTorch, IsolationForest, CUSUM, Mahalanobis, A/B Testing, Hypothesis Testing, Time-Series Forecasting (ARIMA, Prophet), Monte Carlo Simulation, SHAP

ACTIVITIES

Cloud Analytics Lead – GDSC, GITAM Hyderabad

Apr 2022 – Apr 2024

- Built GCP and BigQuery pipelines analyzing behavioral data to quantify investment impact on college admission rates.